AI Co-Design of a Reusable Rocket

(1) Overview

Design a conceptual reusable launch vehicle using AI-assisted engineering workflows.

Students work in teams of three to five, supported by an LLM as an engineering copilot, a professor (Prof. Xu) as chief engineer, and graduate students (Yingjie & Jingjie) as subsystem mentors. Rather than pursuing the best rocket, the course focuses on requirements decomposition, multidisciplinary design optimization, trade-off analysis, uncertainty quantification, and engineering communication.

(2) Schedule

Day	Theme	Deliverable
1 (11-Jul)	Mission definition	Mission requirements
2 (12-Jul)	Rocket fundamentals	First-order sizing
3 (13-Jul)	Propulsion system	Engine selection
4 (14-Jul)	Mass budget and advanced materials	Vehicle architecture
5 (15-Jul)	Aerodynamics and trajectory	Flight profile
6 (16-Jul)	Reusability strategy	Recovery concept
7 (17-Jul)	AI-assisted optimization	Design iteration
8 (18-Jul)	Reliability and economics	Cost and risk analysis
9 (19-Jul)	Final system integration	Technical review
10 (20-Jul)	Design competition	Final presentation

(3) Potential outcomes

Students could produce a mission report, CAD sketches, trajectory simulations, mass budgets, cost models, and AI interaction logs, together with an AI-generated engineering notebook that records each design decision and its rationale.

(4) Future extensions

More ambitious extensions include multi-agent engineering AI, simplified digital twins, evolutionary design competitions, and historical benchmarking against vehicles. A focused late-stage extension could investigate how emerging materials and manufacturing technologies, such as carbon-fiber-reinforced composites and 3D-printed components, may improve structural efficiency, manufacturability, and reusability.

The long-term goal is to teach students how human creativity, physical principles, and AI-assisted exploration can be integrated across mechanics, thermodynamics, fluid dynamics, control, optimization, economics, and AI.